

MACHINING : TURNING OPERATIONS

■ Problem 1 (15/Jan/2021 exam)

A steel bar with a diameter of 30 mm and a length of 160 mm is held by means of a self-centering three-jaw chuck and tailstock (dead centre). The workpiece undergoes external cylindrical turning, which brings the diameter to the final value of 24 mm over a length of 90 mm. The machining is performed in two steps: a roughing step and a finishing step. Considering the roughing step, whose depth of cut is equal to 2 mm, determine:

Q1. The values of cutting force and cutting power for the roughing step, considering a value of $k_{c0,4}$ equal to 2150 MPa, a coefficient x of the Kronenberg relationship equal to 0,29 and a value of main entering angle equal to 90° . The recommended feed is 0,15 mm/rev, while the cutting speed is 260 m/min.

Q2. The optimal tool life and the corresponding cutting speed, in case you want to optimize the production time. The tool change time T_t is equal to 120 seconds, while the constants of the Taylor's law are $C = 250$ and $n = 0,122$.



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■ Problem 2 (10/Sep/2021 exam)

A steel bar with a diameter of 100 mm and a length of 300 mm is held by means of a self-centering three-jaw chuck and tailstock (dead centre). The workpiece undergoes external cylindrical turning, which brings the diameter to the final value of 94 mm over a length of 180 mm. The machining is performed in two steps: a roughing step and a finishing step. Considering the roughing step, whose depth of cut is equal to 2,5 mm, determine:

Q3. The values of **cutting force** and **cutting power**, taking into account a value of $k_{c0,4}$ equal to 2100 MPa, a coefficient x of the Kronenberg relationship equal to 0,29 and a value of main entering angle equal to 45° . The recommended feed is 0,15 mm/rev while the cutting speed is 260 m/min.

Q4. The **optimal tool life** and the **corresponding cutting speed** in case you want to optimize the production cost. The tool changing time is equal to 160 seconds, the cutting-edge cost is equal to 1 €, and the operating cost of the machine tool is 180 €/h, while the constants of the Taylor relation are $C = 250$ and $n = 0,122$.

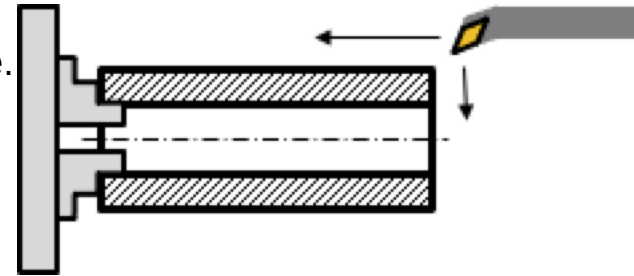


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■ Problem 3 (05/Feb/2021 exam)

The finishing of a hollow cylindrical (tubular) steel component is carried out with the same tool by performing two operations (see figure):

- I. External cylindrical turning over the entire length of the workpiece.
- II. Facing the free base (right).



Using the following data, determine:

1. The cutting speed to maximize the production rate.
2. The maximum allowed feed to meet a surface finish requirement on the outer cylindrical surface equal to $R_a = 0,4 \mu\text{m}$.
3. The minimum clamping pressure to properly hold the workpiece (the feed rate is $0,08 \text{ mm/rev}$, a self-centering three-jaws chuck is used with jaw-workpiece contact area equal to 120 mm^2 and friction coefficient equal to $0,35$).
4. The contact time of the facing operation in the hypothesis of carrying out the machining with a constant number of revolutions per minute equal to 540 and a feed rate of $0,08 \text{ mm/rev}$. And determine the value of the maximum cutting speed in this situation.

Data: Initial external diameter = 236 mm , final external diameter = 233 mm , initial length = 300 mm , internal diameter = 150 mm , $k_{c0,4} = 2100 \text{ MPa}$, $x = 0,29$, tool changing time = 120 s , tool radius = $0,8 \text{ mm}$, $k_r = 100^\circ$, $k_r' = 10^\circ$, Taylor's law coefficient: $C = 510$, $n = 0,35$.

